「Thm.

Let A be a dense subset in a Metric space X.

Every uniformly continuous map $f: A \to \mathbb{R}$ can be extended continuously to X.

Proof. Let $x \in X \setminus A$ be given. Since A is dense in X, given $n \in \mathbb{N}$, $B_{\frac{1}{n}}(x) \cap A \neq \phi$, so we can construct $\{P_n\}$ such that $P_n \in B_{\frac{1}{n}}(x) \cap A$, for all $n \in \mathbb{N}$.

Since f is continuously uniformly on A, a sequence $\{f(P_n)\}$ is Cauchy sequence, because given $\varepsilon > 0$, there is $\delta > 0$ such that $\forall x, y \in A$, $d(x,y) < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$.
And, since $\{P_n\}$ is convergent sequence by the construction, so we can take $N \in \mathbb{N}$ such that $m > n \geq N$ implies $d(P_m, P_n) < \delta$. So

$$m > n \geq N \Rightarrow d(P_m, P_n) < \delta \Rightarrow |f(P_m) - f(P_n)| < \varepsilon.$$

Now, since the $\mathbb{R}$ is complete, $\{f(P_n)\}$ converges for some $L_1 \in \mathbb{R}$.
Meanwhile, let $\{q_n\}$ be a sequence in A which converges to x.
Then, similarly, $\{f(q_n)\}$ converges for some $L_2 \in \mathbb{R}$. To show that $L_1 = L_2$, let $\varepsilon > 0$ be given. By the uniformly continuity, there exists $\delta > 0$ such that

$$\text{for all } x, y \in A,\ |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

Take $N_i \in \mathbb{N}$ $(i = 1,2,3,4)$ satisfying

$$\begin{cases} n \geq N_1 \Rightarrow |f(P_n) - L_1| < \frac{\varepsilon}{3} \\ n \geq N_2 \Rightarrow |f(q_n) - L_2| < \frac{\varepsilon}{3}, \end{cases} \quad \text{and} \quad \begin{cases} n \geq N_3 \Rightarrow d(P_n, x) < \frac{\delta}{2}, \\ n \geq N_4 \Rightarrow d(q_n, x) < \frac{\delta}{2} \end{cases}$$

Now, for $n \geq \max\{N_1, N_2, N_3, N_4\}$, $d(P_n, q_n) \leq d(P_n, x) + d(x, q_n) < \delta$ so that

$$|L_1 - L_2| \leq |L_1 - f(P_n)| + |f(P_n) - f(q_n)| + |f(q_n) - L_2| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Since ε was arbitrary, $L_1 = L_2$. Now, for each $x \in X$, define

$$\bar{f}(x) = \begin{cases} f(x) & \text{if } x \in A \\ \lim_{n\to\infty} f(P_n) & \text{if } x \in X \setminus A, \text{ where } P_n \in A,\ P_n \to x. \end{cases}$$

It is well-defined since the argument above, and continuous on X because given $\varepsilon > 0$, take $\delta > 0$ such that $a, b \in A$, $d(a,b) < \delta \Rightarrow |f(a) - f(b)| < \varepsilon/2$
Given $x, y \in X$ such that $d(x,y) < \frac{\delta}{2}$, take $a_n, b_n \in A$ such that $a_n \to x$, $b_n \to y$.
Then, $|\bar{f}(x) - \bar{f}(y)| \leq |\bar{f}(x) - f(a_n)| + |f(a_n) - f(b_n)| + |\bar{f}(b_n) - y| < \varepsilon$ for large enough $n \in \mathbb{N}$, since $f(a_n) \to \bar{f}(x)$ and $f(b_n) \to \bar{f}(y)$. So, $\bar{f}$ is continuous uniformly. ☑